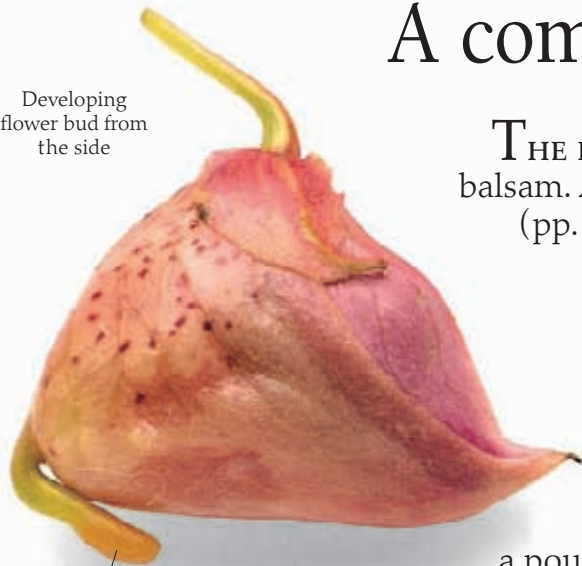


A complex flower

Developing flower bud from the side



Developing nectar spur



Flowers develop in clusters and open one at a time

Pouch-shaped third sepal

Nectar spur



THE FLOWERS ON THESE TWO PAGES come from the Himalayan balsam. Although they have the same basic parts as the lily flower (pp. 16–17), evolution has modified them in different ways so that the two flowers look completely different. Compared with the lily flower, the flowers of the balsam are much more complex and specialized. Himalayan balsam flowers are pollinated by long-tongued insects, such as bees, and they are shaped to ensure that an insect picks up the grains of pollen from the anthers when it approaches and enters the flower. Bees are attracted to the flower by sugary nectar, which is produced in a spur attached to a pouch at the back of the flower. To reach this nectar, a visiting bee first has to land on a platform made up of petals. It then has to climb right inside the flower and stretch out its long tongue. When the bee is in this position, its back touches the anthers. These give it a dusting of pollen, which is carried to the next flower the bee visits.

Stigma Style



Anther

A PAT ON THE BACK

A passion flower has three stigmas, unlike the lily, which has just one. Insects seek out the nectar produced in the base of the passion flower. When the passion flower opens, the anthers, which are lower than the styles (p. 17), dust pollen on to a visiting insect's back. A few hours later, the styles curve downward so that the stigmas are lower than the empty anthers. This enables the stigmas to collect pollen from another insect's back.

Two petals joined together

Fully formed flower from the side

ATTRACTIVE TO INSECTS

Himalayan balsam flowers each have three sepals and five petals, and are symmetrical in as far as each half is a mirror image of the other. Two of the sepals are small flaps at the base of the flower. They protect the young flower bud. During millions of years of evolution, the third sepal has become much larger and is shaped like a pouch. At the end of this pouch lies the narrow spur that produces nectar.